

ENUNCIADO — PROBLEMA 1

Numerical Analysis Preliminary Examination questions
August 2017

Instructions: Closed book. NO CALCULATORS are allowed. This examination contains **seven** problems, each worth 20 points. Do any **five** of the seven problems. Show your work. Clearly indicate which **five** are to be graded.

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1. Consider a differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$ and let $x^* = fl(x)$ be the floating point approximation of x .

(a) Show that the relative error in function evaluation is given by:

$$\left| \frac{f(x) - f(x^*)}{f(x)} \right| \approx \kappa_f(x^*) \left| \frac{x - x^*}{x} \right|$$

where the *condition number* $\kappa_f(x^*) := \left| \frac{xf'(x^*)}{f(x)} \right|$.

- (b) Use the definition of condition number in part (a) to show that the condition number of the product $f(x) \cdot g(x)$ of functions satisfies:

$$\kappa_{fg}(x) \leq \kappa_f(x) + \kappa_g(x).$$

2. Suppose that a function g maps the interval $[a, b]$ into itself and g satisfies a Lipschitz condition with a Lipschitz constant $0 \leq \lambda < 1$, then

(a) Show that the sequence $x_{n+1} = g(x_n)$, $n = 0, 1, \dots$ (for any initial approximation $x_0 \in [a, b]$) converges to a unique fixed point ξ in $[a, b]$.

(b) Moreover, prove the following error bound involved in using the sequence x_n to approximate ξ given by: $|x_n - \xi| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \quad \forall n \geq 1$.

3. Prove that Jacobi method for solving the linear system of equations $Ax = b$ with strictly diagonally dominant matrix A converges with a linear rate.

4. (a) Explain how to find the natural cubic spline for n data points $(x_1, y_1), \dots, (x_n, y_n)$ such that the resulting linear system $Ax = b$ has a matrix A of size n -by- n .

(b) What is the largest number of nonzero elements in A ?

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SOLUCIÓN — PROBLEMA 1

Problema 1 — Número de condición

Sea $f : \mathbb{R} \rightarrow \mathbb{R}$ diferenciable, $x^* = f(x)$.

(a) Fórmula del error relativo en la evaluación de f

Por el teorema del valor medio (o linealización de Taylor de primer orden) alrededor de x^* :

$$f(x) - f(x^*) \approx f'(x^*)(x - x^*)$$

para x cercano a x^* . Dividiendo entre $f(x)$:

$$\frac{f(x) - f(x^*)}{f(x)} \approx \frac{f'(x^*)(x - x^*)}{f(x)} = \frac{x f'(x^*)}{f(x)} \cdot \frac{x - x^*}{x}.$$

Tomando valor absoluto:

$$\left| \frac{f(x) - f(x^*)}{f(x)} \right| \approx \left| \frac{x f'(x^*)}{f(x)} \right| \cdot \left| \frac{x - x^*}{x} \right| = \kappa_f(x^*) \left| \frac{x - x^*}{x} \right|, \quad \kappa_f(x^*) := \left| \frac{x f'(x^*)}{f(x)} \right|. \blacksquare$$

El número de condición $\kappa_f(x^*)$ mide el **factor de amplificación**: convierte el error relativo de la entrada, $|x - x^*|/|x|$ (p. ej. proveniente de la representación de punto flotante), en el error relativo resultante en la salida $f(x)$. Si $\kappa_f \gg 1$, incluso una pequeña perturbación en x puede producir un gran error relativo en $f(x)$: el problema de evaluar f está **mal condicionado**.

(b) Número de condición de un producto

Sea $h(x) = f(x)g(x)$. Por la regla del producto: $h'(x) = f'(x)g(x) + f(x)g'(x)$.

$$\kappa_{fg}(x) = \left| \frac{x h'(x)}{h(x)} \right| = \left| \frac{x[f'(x)g(x) + f(x)g'(x)]}{f(x)g(x)} \right| = \left| \frac{x f'(x)}{f(x)} + \frac{x g'(x)}{g(x)} \right|.$$

Por la desigualdad triangular:

$$\kappa_{fg}(x) \leq \left| \frac{x f'(x)}{f(x)} \right| + \left| \frac{x g'(x)}{g(x)} \right| = \kappa_f(x) + \kappa_g(x). \blacksquare$$

Verificación: si $f = g$ (mismo condicionamiento), $\kappa_{f^2} \leq 2\kappa_f$, consistente con que $h = f^2 \Rightarrow h'/h = 2f'/f$ exactamente (igualdad, no solo desigualdad, cuando $f = g$ y no hay cancelación de signos).

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3. Prove that Jacobi method for solving the linear system of equations $Ax = b$ with strictly diagonally dominant matrix A converges with a linear rate.

4. (a) Explain how to find the natural cubic spline for n data points $(x_1, y_1), \dots, (x_n, y_n)$ such that the resulting linear system $Ax = b$ has a matrix A of size n -by- n .
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Problema 2 — Teorema del punto fijo de Banach en $[a, b]$

(Enunciado idéntico al Problema 3 del examen de enero de 2016; se repite la demostración completa por autocontención.)

$g : [a, b] \rightarrow [a, b]$, Lipschitz con constante $0 \leq \lambda < 1$.

(a) Existencia, unicidad y convergencia

Cota telescópica. $|x_{n+1} - x_n| = |g(x_n) - g(x_{n-1})| \leq \lambda |x_n - x_{n-1}|$, y por inducción $|x_{n+1} - x_n| \leq \lambda^n |x_1 - x_0|$.

Cauchy. Para $m > n$:

$$|x_m - x_n| \leq \sum_{k=n}^{m-1} |x_{k+1} - x_k| \leq |x_1 - x_0| \sum_{k=n}^{m-1} \lambda^k \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0| \xrightarrow{n \rightarrow \infty} 0.$$

Luego (x_n) es de Cauchy en el compacto (completo) $[a, b]$, así que $x_n \rightarrow \xi \in [a, b]$.

Punto fijo. g continua (por ser Lipschitz); tomando límite en $x_{n+1} = g(x_n)$: $\xi = g(\xi)$.

Unicidad. Si ξ_1, ξ_2 son puntos fijos: $|\xi_1 - \xi_2| = |g(\xi_1) - g(\xi_2)| \leq \lambda |\xi_1 - \xi_2| \xrightarrow{\lambda < 1} 0 \Rightarrow \xi_1 = \xi_2$ (pues $\lambda < 1$). ■

(b) Cota de error

De la desigualdad de Cauchy anterior, fijando n y tomando $m \rightarrow \infty$:

$$|x_n - \xi| = \lim_{m \rightarrow \infty} |x_n - x_m| \leq \frac{\lambda^n}{1 - \lambda} |x_1 - x_0|, \quad \forall n \geq 1. \quad \blacksquare$$

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Problema 3 — Convergencia lineal de Jacobi bajo dominancia diagonal

Sea A estrictamente diagonalmente dominante por filas: $r := \max_i \frac{1}{|a_{ii}|} \sum_{j \neq i} |a_{ij}| < 1$. Iteración de Jacobi: $x^{(k+1)} = B_J x^{(k)} + D^{-1}b$, con $(B_J)_{ij} = -a_{ij}/a_{ii}$ ($j \neq i$), $(B_J)_{ii} = 0$.

Paso 1 — cota de la norma de B_J .

$$\|B_J\|_\infty = \max_i \sum_{j \neq i} \frac{|a_{ij}|}{|a_{ii}|} = r < 1.$$

Paso 2 — convergencia. Sea x^* el punto fijo, $e^{(k)} = x^{(k)} - x^*$. Restando la iteración de la ecuación de punto fijo: $e^{(k+1)} = B_J e^{(k)}$, y por inducción $e^{(k)} = B_J^k e^{(0)}$. Por submultiplicatividad:

$$\|e^{(k)}\|_\infty \leq \|B_J\|_\infty^k \|e^{(0)}\|_\infty = r^k \|e^{(0)}\|_\infty \xrightarrow{k \rightarrow \infty} 0$$

(pues $0 \leq r < 1$): converge para todo $x^{(0)}$.

Paso 3 — la tasa es lineal. Por definición, una sucesión converge **linealmente** con razón ρ si $\|e^{(k+1)}\|/\|e^{(k)}\|$ está acotada (asintóticamente) por una constante $\rho \in (0, 1)$. Aquí se tiene, directamente de $e^{(k+1)} = B_J e^{(k)}$ y la submultiplicatividad de la norma:

$$\frac{\|e^{(k+1)}\|_\infty}{\|e^{(k)}\|_\infty} \leq \|B_J\|_\infty = r < 1 \quad \text{para todo } k$$

(cuando $e^{(k)} \neq 0$), es decir, en cada paso el error se reduce por un factor **a lo sumo** r , uniformemente en k . Esto es exactamente convergencia geométrica/lineal con razón (a lo sumo) r : la sucesión de errores está dominada por una progresión geométrica $r^k \|e^{(0)}\|_\infty$, y esta tasa es, en general, **óptima** — tomando $e^{(0)}$ alineado con la dirección que maximiza $\|B_J v\|_\infty / \|v\|_\infty$ se alcanza $\rho(B_J)$, que satisface $\rho(B_J) \leq \|B_J\|_\infty = r$, confirmando que no puede esperarse una tasa mejor que geométrica (lineal) en general — a diferencia de métodos como Newton, que logran convergencia cuadrática. ■

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(c) Repeat steps (a) and (b) for the backward Euler method.
7. (a) Derive the quadrature rule for approximating the integral $\int_a^{a+h} f(x) dx$ using the two nodes $x_1 = a$ and $x_2 = a + 2h/3$ such that the quadrature rule is exact for all polynomials up to the highest possible degree.
(b) Show that the quadrature rule above is exact on all polynomials of degree 2.
(c) Assume that the node $x_1 = a$ is fixed but you are allowed to change the node x_2 . Is $x_2 = a + 2h/3$ the choice for which the quadrature rule is exact for all polynomials up to the highest possible degree? Why or why not?

SOLUCIÓN — PROBLEMA 4**Problema 4 — Spline cúbico natural: sistema lineal $n \times n$**

(a) Construcción

Dados n puntos $(x_1, y_1), \dots, (x_n, y_n)$ con $x_1 < \dots < x_n$, el spline cúbico natural $S(x)$ es una función cúbica a trozos S_i en cada $[x_i, x_{i+1}]$ ($i = 1, \dots, n-1$, $h_i = x_{i+1} - x_i$) tal que:

- **Interpolación:** $S_i(x_i) = y_i$, $S_i(x_{i+1}) = y_{i+1}$.
- **Continuidad C^1 :** $S'_i(x_{i+1}) = S'_{i+1}(x_{i+1})$.
- **Continuidad C^2 :** $S''_i(x_{i+1}) = S''_{i+1}(x_{i+1})$.
- **Condición natural:** $S''(x_1) = 0$, $S''(x_n) = 0$.

Parametrizando por los "momentos" $M_i := S''(x_i)$, $i = 1, \dots, n$, la continuidad de S' en los nodos interiores produce, para $i = 2, \dots, n-1$, el sistema tridiagonal clásico:

$$h_{i-1}M_{i-1} + 2(h_{i-1} + h_i)M_i + h_iM_{i+1} = 6 \left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}} \right),$$

junto con las dos ecuaciones triviales de frontera $M_1 = 0$, $M_n = 0$. Esto da el sistema $AM = \mathbf{b}$ de tamaño $n \times n$ ($\mathbf{M} = (M_1, \dots, M_n)^T$):

$$A = \begin{pmatrix} 1 & 0 & & & & \\ h_1 & 2(h_1 + h_2) & h_2 & & & \\ & h_2 & 2(h_2 + h_3) & h_3 & & \\ & & \ddots & \ddots & \ddots & \\ & & & h_{n-2} & 2(h_{n-2} + h_{n-1}) & h_{n-1} \\ & & & & 0 & 1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ \vdots \\ 6 \left(\frac{y_{i+1} - y_i}{h_i} - \frac{y_i - y_{i-1}}{h_{i-1}} \right) \\ \vdots \\ 0 \end{pmatrix}.$$

Una vez resuelto $\mathbf{M}$, los coeficientes de cada pieza cúbica se recuperan con la fórmula estándar

$$S_i(x) = \frac{M_i}{6h_i}(x_{i+1} - x)^3 + \frac{M_{i+1}}{6h_i}(x - x_i)^3 + \left(\frac{y_i}{h_i} - \frac{M_i h_i}{6} \right)(x_{i+1} - x) + \left(\frac{y_{i+1}}{h_i} - \frac{M_{i+1} h_i}{6} \right)(x - x_i).$$

(b) Máximo número de elementos no nulos de A

A es **tridiagonal**: cada fila tiene a lo sumo 3 entradas no nulas (diagonal y las dos adyacentes), salvo las dos filas de frontera (fila 1 y fila n), que solo tienen 1 entrada no nula cada una (el 1 de $M_1 = 0$, $M_n = 0$). El conteo exacto para esta construcción es

$$1 + 3(n-2) + 1 = 3n - 4$$

no nulos. (En general, cualquier matriz tridiagonal $n \times n$ tiene a lo sumo $3n - 2$ entradas no nulas; aquí, por las dos filas triviales de frontera, el conteo real es ligeramente menor: $3n - 4$.)

(c) Método numérico apropiado

Como A es tridiagonal (y su bloque interior $(n-2) \times (n-2)$ es simétrico, diagonalmente dominante y definido positivo, pues $2(h_{i-1} + h_i) > h_{i-1} + h_i$), el método apropiado es el

algoritmo de Thomas (eliminación gaussiana especializada para matrices tridiagonales: barrido de eliminación hacia adelante seguido de sustitución hacia atrás), que resuelve el sistema en $O(n)$ operaciones en lugar de las $O(n^3)$ de una eliminación gaussiana densa genérica. Alternativamente, al ser el bloque interior simétrico definido positivo, puede usarse una **factorización de Cholesky tridiagonal** (LDL^T), también de costo $O(n)$.

ENUNCIADO — PROBLEMA 5

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Problema 5 — Aproximación de segundo orden de $f''(x)$

(a) Fórmula

$$f''(x) \approx \frac{f(x-h) - 2f(x) + f(x+h)}{h^2}.$$

(b) Análisis del error (con $f \in C^4$)

Por Taylor con residuo de Lagrange de cuarto orden:

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2}f''(x) + \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(\xi_1), \quad \xi_1 \in (x, x+h),$$

$$f(x-h) = f(x) - hf'(x) + \frac{h^2}{2}f''(x) - \frac{h^3}{6}f'''(x) + \frac{h^4}{24}f^{(4)}(\xi_2), \quad \xi_2 \in (x-h, x).$$

Sumando (los términos impares en h se cancelan):

$$f(x+h) + f(x-h) = 2f(x) + h^2f''(x) + \frac{h^4}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_2)].$$

Despejando:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{24}[f^{(4)}(\xi_1) + f^{(4)}(\xi_2)].$$

Si $f^{(4)}$ es continua en $[x-h, x+h]$, por el teorema del valor intermedio el promedio $\frac{1}{2}[f^{(4)}(\xi_1) + f^{(4)}(\xi_2)] = f^{(4)}(\xi)$ para algún $\xi \in [x-h, x+h]$, dando la forma clásica más precisa:

$$\frac{f(x+h) - 2f(x) + f(x-h)}{h^2} = f''(x) + \frac{h^2}{12}f^{(4)}(\xi).$$

Con $M_4 := \max_{[x-h, x+h]} |f^{(4)}|$:

$$\left| \frac{f(x+h) - 2f(x) + f(x-h)}{h^2} - f''(x) \right| \leq \frac{M_4}{12}h^2 = O(h^2).$$

Segundo orden de exactitud confirmado. ■

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Problema 6 — Estabilidad de Euler explícito e implícito

$x' = \lambda x$, $x(0) = 1$, paso h .

(a)–(b) Euler explícito

$$x_{k+1} = x_k + h\lambda x_k = (1 + h\lambda)x_k.$$

Estabilidad ($|x_{k+1}|$ no crece): $|1 + h\lambda| \leq 1$. Para $\lambda \in \mathbb{R}$, $\lambda < 0$: esto equivale a $-2 \leq h\lambda \leq 0$, es decir

$$0 < h \leq \frac{2}{|\lambda|}.$$

(Para λ complejo, la región de estabilidad absoluta es el disco $|1 + z| \leq 1$ en el plano $z = h\lambda$, un disco de radio 1 centrado en -1 .)

(c) Euler implícito (hacia atrás)

$$x_{k+1} = x_k + h\lambda x_{k+1} \implies (1 - h\lambda)x_{k+1} = x_k \implies x_{k+1} = \frac{x_k}{1 - h\lambda}.$$

Estabilidad: $\left| \frac{1}{1 - h\lambda} \right| \leq 1 \iff |1 - h\lambda| \geq 1$. Para $\lambda < 0$ real y $h > 0$: $1 - h\lambda = 1 + h|\lambda| > 1$ **siempre**, así que la condición se satisface para **todo** $h > 0$: el método es **incondicionalmente estable** (A-estable) en este caso, sin restricción alguna sobre el tamaño de paso. (La región de estabilidad de Euler implícito es el exterior del disco $|1 - z| \geq 1$, que contiene todo el semiplano izquierdo $\operatorname{Re}(\lambda) \leq 0$: de ahí su A-estabilidad.)

Verificación cualitativa: para $\lambda < 0$ la solución exacta $x(t) = e^{\lambda t}$ $\rightarrow 0$ decae; Euler implícito reproduce esa decaimiento para cualquier h (factor $|R(z)| = 1/|1 - z| < 1$ siempre que $\operatorname{Re}(z) < 0$), mientras que Euler explícito solo lo reproduce si h es suficientemente pequeño ($h \leq 2/|\lambda|$) — la diferencia esencial entre esquemas explícitos e implícitos frente a la rigidez (*stiffness*).

ENUNCIADO — PROBLEMA 7

5. (a) Give a formula for a second order approximation of the second derivative $f''(x)$.
(b) Assume that f is four times continuously differentiable in a neighborhood of x and show that your approximation in part (a) is second order by deriving an appropriate upper bound on the error.
6. Consider applying the forward Euler method to solve the initial value problem, $x' = \lambda x$ with $x(0) = 1$ using a step size $h = t_{k+1} - t_k$.
(a) Write the formula for $x_{k+1} = x(t_{k+1})$ in terms of $x_k = x(t_k)$.
(b) For which values of h and λ will this method be stable?
(c) Repeat steps (a) and (b) for the backward Euler method.
7. (a) Derive the quadrature rule for approximating the integral $\int_a^{a+h} f(x) dx$ using the two nodes $x_1 = a$ and $x_2 = a + 2h/3$ such that the quadrature rule is exact for all polynomials up to the highest possible degree.
(b) Show that the quadrature rule above is exact on all polynomials of degree 2.
(c) Assume that the node $x_1 = a$ is fixed but you are allowed to change the node x_2 . Is $x_2 = a + 2h/3$ the choice for which the quadrature rule is exact for all polynomials up to the highest possible degree? Why or why not?

Problema 7 — Cuadratura de 2 nodos con un nodo fijo (Gauss-Radau)

$$\int_a^{a+h} f(x) dx \approx w_1 f(a) + w_2 f\left(a + \frac{2h}{3}\right).$$

(a) Derivación de los pesos

Cambio de variable $u = (x - a)/h \in [0, 1]$, $x = a + uh$, $dx = h du$; sea $\tilde{f}(u) = f(a + uh)$, nodos en $u = 0$ y $u = 2/3$. Se busca $h \int_0^1 \tilde{f} du = w_1 \tilde{f}(0) + w_2 \tilde{f}(2/3)$ exacta para $\tilde{f} = 1, u$:

$$\tilde{f} = 1 : \quad h = w_1 + w_2.$$

$$\tilde{f} = u : \quad \frac{h}{2} = w_2 \cdot \frac{2}{3} \implies w_2 = \frac{3h}{4}.$$

De la primera: $w_1 = h - \frac{3h}{4} = \frac{h}{4}$.

$$w_1 = \frac{h}{4}, \quad w_2 = \frac{3h}{4}, \quad \int_a^{a+h} f(x) dx \approx \frac{h}{4} f(a) + \frac{3h}{4} f\left(a + \frac{2h}{3}\right).$$

(b) Exactitud para polinomios de grado 2

Verificamos $\tilde{f} = u^2$ (equivalente, tras el cambio de variable afín, a verificar $f = x^2$, pues transformaciones afines preservan el grado polinómico):

$$\text{LHS: } h \int_0^1 u^2 du = \frac{h}{3}, \quad \text{RHS: } w_2 \left(\frac{2}{3}\right)^2 = \frac{3h}{4} \cdot \frac{4}{9} = \frac{3h \cdot 4}{36} = \frac{h}{3}.$$

Coinciden exactamente: la regla es exacta para u^2 (y, por combinación lineal con los casos 1, u ya verificados, para **todo** polinomio de grado ≤ 2). ■

(Comprobación de que no llega a grado 3: con $\tilde{f} = u^3$, LHS = $h/4$, RHS = $w_2(2/3)^3 = \frac{3h}{4} \cdot \frac{8}{27} = \frac{2h}{9} \neq \frac{h}{4}$: falla en grado 3, así que el grado máximo de exactitud de esta regla es exactamente 2.)

(c) ¿Es $x_2 = a + 2h/3$ la elección óptima del segundo nodo?

Sí. Esta es precisamente la construcción de una **cuadratura de Gauss-Radau: una regla de n nodos en la que un** nodo se fija de antemano en un extremo del intervalo (aquí $x_1 = a$), y los $n - 1$ nodos restantes (y todos los n pesos) se eligen para maximizar el grado de exactitud. La teoría de Gauss-Radau establece que el grado máximo alcanzable con n nodos (uno de ellos fijo en un extremo) es $2n - 2$; aquí $n = 2$, así que el techo teórico es $2n - 2 = 2$ — exactamente el grado obtenido en (b), y **ningún otro valor de x_2** (con $x_1 = a$ fijo) puede superar el grado 2.

Más aún, la elección específica $x_2 = a + 2h/3$ (es decir $\theta = 2/3$ en la variable normalizada u) es precisamente la **raíz de Radau** asociada al peso constante en $[0, 1]$ con nodo fijo en $u = 0$: en la

formulación estándar de Gauss-Radau en $[-1, 1]$ con dos nodos y uno fijo en -1 , los nodos son -1 y $1/3$, con pesos $1/2$ y $3/2$; bajo el cambio de variable afín $u = (t + 1)/2$ que lleva $[-1, 1] \rightarrow [0, 1]$, el nodo $t = 1/3$ se transforma en $u = (1/3 + 1)/2 = 2/3$ — coincidiendo exactamente con $\theta = 2/3$. Por lo tanto, $x_2 = a + 2h/3$ **sí es la elección óptima**: cualquier otra posición de x_2 (distinta de la raíz de Radau) solo alcanzaría, genéricamente, exactitud de grado 1 (dos condiciones para dos pesos libres), mientras que esta elección particular consigue el grado máximo teórico, 2, gracias a la ortogonalidad del polinomio $u - 2/3$ respecto al peso constante en $[0, 1]$ tras imponer el nodo fijo en $u = 0$. ■